

**QUEUING THEORY FOR HEALTHCARE OPERATIONS
MANAGEMENT: A Case Study of University of Benin Health Center and
Faith Mediplex.**

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ABSTRACT

Queues are characterized structures fashioned to maintain order and create a hold on time, money and human contribution towards development and efficient performance of any system. Queues are experienced in our everyday activities. Queue causes inconvenience to individuals (patients) and economic costs to firms and organizations. Patients wait for minutes, hours, days or months to receive medical service-waiting before, during or after being served. Queuing theory is a mathematical approach to the study of waiting in lines/queues. This research presents the results of a study that evaluates the effectiveness of a queuing model in identifying the major bottleneck in healthcare operations. It uses chi-square and *Erlang's queuing* to analyse data collected from the University of Benin Health Center and the Faith Mediplex center over one month period. Results showed that queue characteristics existing at the healthcare centers during the situation analysis was a single server multiple queue model. However, after the study was done involving staff at the understudy healthcare centers, it was discovered that queuing is mainly found in record unit and doctor consultation waiting lobby. We finally came to a conclusion and made the some recommendations on how best healthcare centers can maximize the benefits of queuing model for good and effective operations.

KEYWORDS: Operation Management, Healthcare, Queuing Theory,

INTRODUCTION

Customer satisfaction has become a serious concern in service sector. On Healthcare industry, a number of initiatives have been introduced to enhance customer satisfaction. The healthcare industry providers globally are experiencing increasing pressure to concurrently reduce cost and improve the access and quality of care they deliver. Many healthcare institutions are confronted with long waiting times, delays, and queues of patients. Long waiting time in any hospital is considered as an indicator of poor quality and needs improvement. Managing waiting lines create a great dilemma for managers seeking to improve the return on investment of their operations. Customers also dislike waiting for long time. If the waiting time and service time is high customers may leave the queue prematurely and this in turn results in customer dissatisfaction. This will reduce customer demand and eventually revenue and profit.

Queues are form when entities that request service, typically referred to as customers, arrive at a service facility and cannot be served immediately upon arrival. In healthcare delivery systems, patients are typically the customers and either outpatient clinics or diagnostic imaging centers or hospitals are the service facilities. A common feature of the vast majority of queuing models is that customers are discrete, and the number of customers waiting in the service facility is integer valued.

The organizations that care for persons who are ill and injured vary widely in scope and scale, from specialized outpatient clinics to large, urban hospitals to regional healthcare systems. Despite these differences, one can view the healthcare processes that these organizations provide as queuing systems in which patients arrive, wait for service, obtain service, and then depart. The healthcare processes also vary in complexity and scope, but they all consist of a set of activities and procedures (both medical and non-medical) that the patient must undergo in order to receive the needed treatment. The resources (or servers) in these queuing systems are the trained personnel and specialized equipment that these activities and procedures require.

Effectively managing patient flow in an outpatient unit is a key to achieving operational excellence as well as ensuring clinical quality. It is especially so for an outpatient department in a large Health center as it handles very large volume of patients with a diverse case mix.

Although the number of health care centers are large but there are still many people who goes for treatment in overseas, especially the upper classes. The Healthcare industry must consider what makes some people prefer to seek treatment in overseas.

Statement of Research Problem

Today's healthcare is working under great pressure. Along with improved medical and healthcare science and possibly healthier lifestyles the proportion of elderly in the population continuously increases. The expectations on healthcare delivery are increasing with enhanced medical care, improved diagnosis techniques and efficiency of treatments. This evidently conveys a general increased demand for healthcare and tends to raise healthcare costs. Consequently the importance of resource planning and efficiency analysis to assist healthcare decision makers to control cost development has increased simultaneously.

There is a need for more realistic representation of outpatient clinics. Effective appointment systems have the goal of matching demand with capacity so that resources are better utilized and patient waiting times are minimized. Patient waiting times and waiting-room congestion are two of the few tangible quality elements. Well-designed appointment systems have the potential to increase the utilization of expensive personnel and equipment-based medical resources as well as reducing waiting times for patients.

This study is therefore required to understand the expected waiting time of customers and the actual waiting time in the health centers, where, the gap between the actual and expected waiting time can be analyzed to know any improvements are required.

Research Questions

Five dominant questions being reviewed by this research include;

1. How does the model of healthcare, service the arrival pattern of patient in healthcare centers?
2. Is an existing outpatient service already fulfilling the minimum service standard of health centers?

3. If they increase or decrease the number of staffs or top management staffs, how would this effect patient waiting time, the length of a medical treatment and the total time spent in clinic by patient?
4. What factors are causing high outpatient waiting times?
5. What advantages can we obtain with a queuing approach in order to evaluate on the effects of government policies related to healthcare centers waiting times?

Objectives of the Study

The objectives of this research study include the following:

1. To determine the major bottleneck in the system of healthcare operations management.
2. To find out why queuing theory is needed in healthcare operations management.
3. To identify hotspots that may be missed by other theories in healthcare operations.

Hypotheses

The following hypotheses is to be tested:

Null Hypothesis: H_0 : There is no significant difference between the service rates the patients expect and the actual service rates followed by different departments.

Alternative Hypothesis: H_1 : There is a significant difference between the service rates the patients expect and the actual service rates followed by different departments.

Relevance of the Study

This research project allows the healthcare industry to notice what impact has the improvement do in the healthcare performance. Moreover, this study will show what factors can be improved in the reducing patient waiting time. This research expected to bring new ideas and concept to be implemented in healthcare operations in Nigeria, especially public health centers since the numbers of researches about this field are limited.

REVIEW OF RELATED LITERATURE

Many organizations, such as banks, airlines, telecommunications companies, and police departments, routinely use queuing models to help determine capacity levels needed to respond to experienced demands in a timely fashion. Long waiting time in any hospital is considered as an indicator of poor quality and needs improvement. Managing waiting lines create a great dilemma for managers seeking to improve the return on investment of their operations. Customers also dislike waiting for long time. If the waiting time and service time is high customers may leave the queue prematurely and this in turn results in customer dissatisfaction. This will reduce customer demand and eventually revenue and profit (Biju, M. K.; Naeema, K. and Faisal, U. 2011). Though queuing analysis has been used in hospitals and other healthcare settings, its use in this sector is not widespread. With rapid change and realignment of healthcare system, new lines of services and facilities to render the same, severe financial pressure on the healthcare organizations, and extensive use of expanded managerial skills in healthcare setting, use of queuing models has become quite prevalent in it.

In an era of healthcare reform, improving quality and safety, and decreasing healthcare cost have become even more important goals than before. With rapid change and realignment of healthcare system, new lines of services and facilities to render the same, severe financial pressure on the healthcare organizations, and extensive use of expanded managerial skills in healthcare setting, the use of queuing model has become a prevalent analytical tool (Singh, 2007). Scientific management of patient flow is at the heart of our ability to achieve these goals. While on one hand we are faced with overcrowded facilities, on the other hand, the industry's financial conditions do not allow us to add resources liberally. One key challenge is our ability to match

random patient demand to fixed capacity. Queuing theory is a methodology that addresses this very challenge. Queuing theory was first used in telecommunications and then was adopted by all major industries, like airlines, the Internet and most service-delivery organizations. In the health care industry, however, queuing theory has not been utilized until recently. When used appropriately, the results are often dramatic: saving time, increasing revenue, and increasing staff and patient satisfaction.

Conceptual Framework

The literature review consists of three portions. A review of the basics of traditional queuing models and queuing network models, based primarily on the following references: Fundamentals of Queuing Theory (Gross, G. and Harris, C., 1998), An Introduction to Queuing Networks (Jean Walrand, 1988), Introduction to Operations Research (Hiller, H. S. and Lieberman, G. L., 1967), Principles of Operations Research (Wagner, H. M., 1969) and Stochastic Modeling: Analysis and Simulation (Barry L. Nelson, 1995).

Queuing theory, the mathematical study of waiting in lines, is a branch of operations research because the results often are used when making business decisions about the resources needed to provide service. At its most basic level, queuing theory involves arrivals at a facility (i.e., computer store, pharmacy, bank) and service requirements of that facility (i.e., technicians, pharmacists, tellers). The number of arrivals generally fluctuates over the course of the hours that the facility is available for business. The theory enables mathematical analysis of several related processes, including arriving at the (back of the) queue, waiting in the queue (essentially a storage process), and being served at the front of the queue. The theory permits the derivation

and calculation of several performance measures including the average waiting time in the queue or the system, the expected number waiting or receiving service, and the probability of encountering the system in certain states, such as empty, full, having an available server or having to wait a certain time to be served.

Queuing management consists of three major components:

1. How customers arrive
2. How customers are serviced
3. The condition of the customer exiting the system

Arrivals: Arrivals are divided into two types:

1. Constant – exactly the same time period between successive arrivals (i.e., machine controlled).
2. Variable – random arrival distributions, which is a much more common form of arrival.

A good rule of thumb to remember the two distributions is that time between arrivals is exponentially distributed and the numbers of arrivals per unit of time is Poisson distributed.

The Servicing or Queuing System: The servicing or queuing system consists of the line(s) and the available number of servers. Factors to consider include the line length, number of lines and the queue discipline. Queue discipline is the priority rule, or rules, for determining the order of service to customers in a waiting line. One of the most common used priority rules is “first come, first served” (FCFS). Others include a reservations first, treatment via triage (i.e., emergency rooms of health centers), highest-profit customer first, largest orders first, “best” customers first and longest wait-time first. An important feature of the waiting structure is the time the customer

spends with the server once the service has started. This is referred to as the service rate: the capacity of the server in numbers of units per time period (i.e., 15 orders per hour).

Exit: There are two possible outcomes after a customer is served. The customer is either satisfied or not satisfied and requires re-service.

The queue discipline refers to the order in which members of the queue are selected for service (Hillier and Lieberman, 2001). Winston and Albright (1997) posit that the usual queue discipline is first come, first served (FCFS or FIFO), where customers are served in order of arrival. In this study the case hospitals use FCFS queuing discipline. Although, sometimes there are other service disciplines: last come, first served (which happens sometime in case of emergencies), or service-in-random order and priority rule. Davis et al, (2003) assert that reservations first, emergencies first, highest profit customer first, largest orders first, best customers first, longest waiting time in line, and soonest promised date are other examples of queue discipline. Unless otherwise stated, the queuing model adopted in this study assumes arrival from infinite source with infinite queue and with first in first served (FCFS) queue discipline. Obamiro, (2010)

Service Mechanism: According to Mosek and Wilson (2001), service mechanism describes how the customer is served. In a single server system each customer is served by exactly one server, even though there may be multiple servers. In most cases, service times are random and they may vary greatly. Sometimes the service time may be similar for each job or constant. The service mechanism also describes the number of servers. A queuing system may operate with a single server or a number of parallel servers. An arrival who finds more than one free server may choose at random any one of them for receiving service. If he finds all the servers busy, he joins a queue

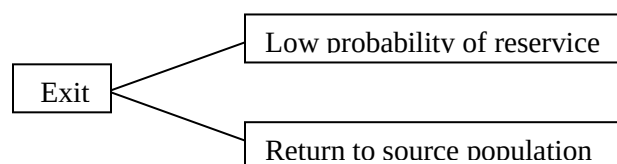
common to all servers. The first customer from the common queue goes to the server who becomes free first (Medhi, 2003).

Capacity of the System: A system may have an infinite capacity-that is, the queue in front of the server(s) may grow to any length. Furthermore, there may be limitation of space and so when the space is filled to capacity, an arrival will not be able to join the system and will be lost to the system. The system is called a delay system or a loss system, according to whether the capacity is infinite or finite respectively (Medhi, 2003).

Departure: Once customers are served, they depart and may not likely re-enter the system to queue again. It is usually assumed that departing customers do not return into the system immediately (Adedayo, et al., 2006). Chase et al.,(2004) is of the opinion that once a customer is served, two exit fates are possible as shown in figure 2.3.

1. The customer may return to the source population and immediately become a competing candidate for service again.
2. There may be a low probability of re-service. In hospitals, departure means home discharge, admission or death (Smith and Mayhew, 2008).

Figure 2.1: Departure



Source: Davis et al., (2005)

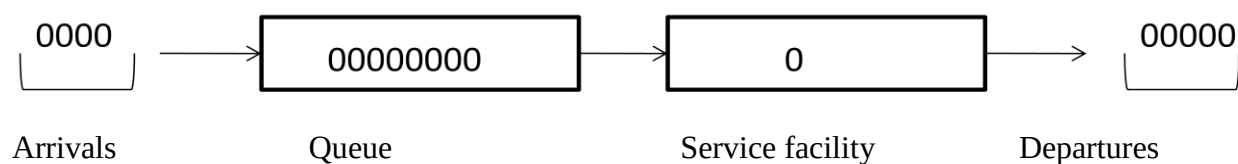
Types of Queuing System

There are four major types of queuing system and different combinations of the same can be adopted for complex networks. Lapin (1981) broadly categorized queuing system structures into the following

1. Single-server, Single-phase system:

Single-phase means only one stop for service.. This is a situation in which single queue of customers are to be served by a single service facility (server) one after the other. An example is flu vaccination camp where a nurse practitioner is the server who does all the work (i.e. .paper work and vaccination (Singh, 2007). Diagrammatically it is depicted in figure 2.2.

Figure 2.2: Single-server, Single phase System.

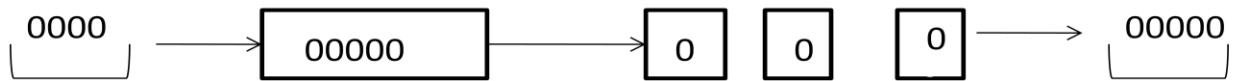


Source: Obamiro, (2010)

2. Single-server, Multiple-phases System:

In this situation, there's still a single queue but customers/patients receive more than one kind of service before departing the queuing system as shown in figure 2.3. For example, at outpatient department, patient first arrive at the registration desk, get the registration done and then wait in a queue to see a nurse for ancillary services before being seen by the consultant (physician). Patients have to join queue at each phase of the system.

Figure 2.3: Single-server, Multiple phases System



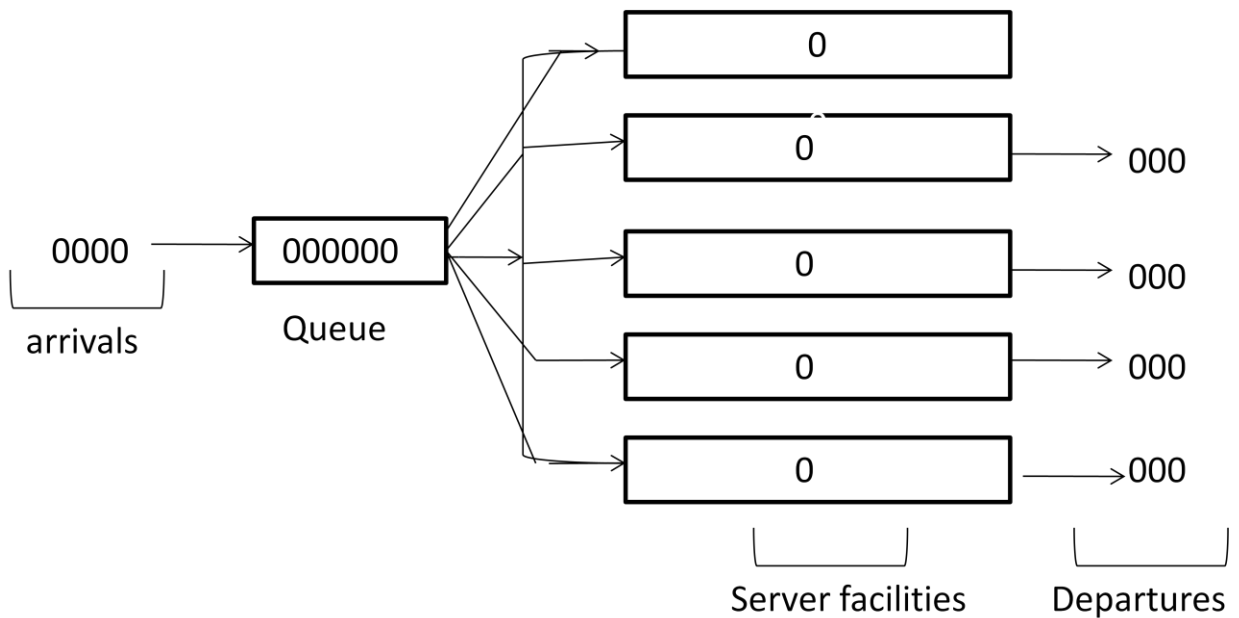
Source: Obamiro, (2010)

3. Multiple-servers, Single-phase System:

This is a queuing system characterized by a situation whereby there is a more than one service facility (servers) providing identical service but drawn on a single waiting line (Obamiro, 2010).

An example is patient waiting to see consultants (physicians) at general outpatient department of teaching hospitals as illustrated by figure 2.4.

Figure 2.4: Multiple-servers, Single phase System

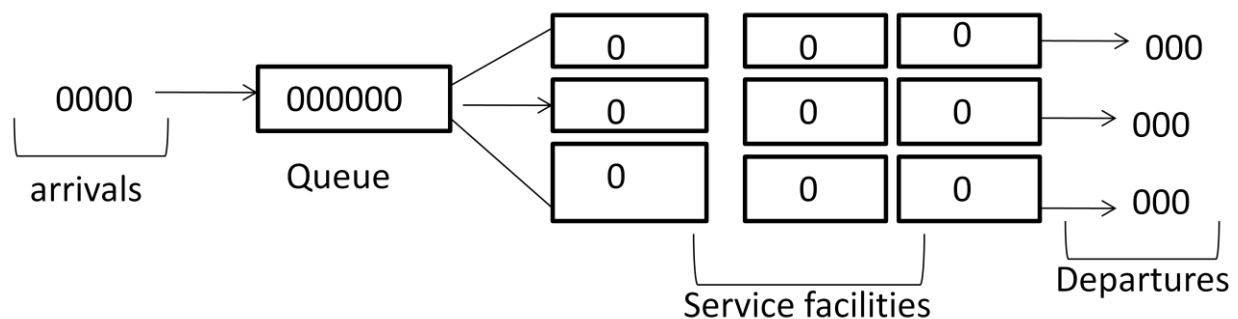


Source: Obamiro, (2010)

4 Multiple servers, Multiple-phases System:

According to Singh (2007), this type of system has numerous queues and a complex network of multiple phases of services involved as can be seen in figure 2.5. This is the type of queuing system adopted in this study. This type of service is typically seen in a hospital setting, multi-specialty outpatient clinics, patient first form the queue for registration, then he/she is triage for assessment, then for diagnostics, review, treatment, intervention or prescription and finally exits from the system or triage to different provider.

Figure 2.5: Multiple-servers, Multiple-phase System



Source: Obamiro, (2010)

Queuing Model Analysis

Queuing models are used to achieve a balance or trade-off between capacity and service delays. Notation for describing the characteristics of a queuing model was first suggested by David G. Kendall in 1953. Kendall's notation introduced an A/B/C queuing notation that can be found in all standard modern works on queuing theory. Two simple single-server models help answer meaningful questions and also address the curse of utilization and the curse of variability. One model assumes variable service time while the other assumes constant service time.

J. Jackson (1957, 1963) made notable contributions to the development of queuing network models. A Jackson model is probably the most researched and widely applied network model in

various fields, including the healthcare field. Jackson's major contribution was to find a "product-form" steady-state solution for in open and closed models with a tandem or a feed-forward flow configuration. In a network model, various numbers of entities can exist at multiple stations, and the state of the system is described by the joint probability distribution for the number of entities at each station. Numerous theoretical works were published which expanded the Jackson model, and many of those examined or modified the Jackson properties (i) and (ii), in particular, in an open model. Among those, Disney (1981) and Melamed (1979) are widely known. Disney (1981) examined the internal arrival rate distribution with feedback flow as a generalization of Jackson model. His research showed that when a system has any kind of feedback flow, the internal flows in the system do not follow the Poisson distribution. Thus, the assumption of Poisson arrival is justified only when the system under consideration has either a tandem or arbitrarily linked network configuration with feed-forward flows. It is, however, known that the Jackson's product-form solution holds regardless of whether or not internal flows are Poisson. Melamed (1979) extended Burke's finding in an open Jackson system. He showed that departure rates from internal stations to outside the system are mutually independent if arriving rates to all internal stations follow the Poisson distribution. The finding, in turn, means that the sum of all departure rates from the network must also be Poisson.

Service Operation in Healthcare Industry

Healthcare service is a patient-oriented service that requires continuous interaction with customers. It utilizes facilities and equipment, and consumes a large volume of nursing care. Therefore, it becomes increasingly important to healthcare executives to understand what kind of facility, equipment, and workforce decisions are critical to achieve the commonly acknowledged

goal of providing quality health service at a reasonable cost. Until now research on operations strategy in Indonesian public hospitals has not been well developed, especially when it relates operations strategy to the current health service condition. Most of the operations-oriented studies focus narrowly on issues of hospital cost containment, capacity planning, or personnel scheduling.

Healthcare management has evolved into a dynamic and complex field. This diverse industry is always changing due to scientific discoveries that bring significant contributions to improve the health standards of our communities. The changes have moved on to determine new ways about how and where healthcare is provided. Li et al. found that the obvious differences in previous hospital research and current hospital practice in managing demand is that previous research tends to focus on a reactive approach to manage demand through internal improvement of facility utilization and better scheduling policies. Integrated Healthcare Management is the systematic application of processes and shared information to optimize the coordination of benefits and care for the healthcare consumer. Integration of healthcare management not only observes the relationship between customer (patients) and hospitals but also the inter-relationship between departments in the healthcare systems. This integrated management will support the effectiveness and efficiency of hospital. Hospital strategic planning can be performed at the corporate level by examining the hospital "system" which would include hospitals, walk-in clinics, and other health-oriented businesses.

Similarities and Differences between Industrial Operations Management and Healthcare Operations Management

Before discussing in detail the applications of operations management to the healthcare industry, it is important to acknowledge the similarities and differences between traditional “industrial” operations management and health care-specific operations management.

In the manufacturing industry, competition has historically been a driving force for evolution in operations management. Competition creates a high pressure on performance in terms of quality, efficiency and flexibility. Production control or logistics can be defined as the coordination of supply, production and distribution processes in manufacturing systems to achieve specific delivery flexibility and delivery reliability at minimum costs (Bertrand et al., 1990). Related objectives are to decrease the lead-times, delivery times and costs and to increase throughput, revenues and profit of the organization. Logistics-oriented manufacturing has contributed in many circumstances to improvements in customer performance (delivery times, delivery reliability) as well as efficiency by the better balancing of delivery performances and efficiency.

Some Challenges Confronted by the Healthcare Industry

Increased complexity of processes by shorter lengths of stay of patients, a shift from inpatient treatment towards ambulatory treatment and day care, use of new technology and increased specialization;

Need for efficient utilization of resources and reduction of costs: first, because treatment is concentrated in a shorter time-space, and second, because of the political pressure to control national health care expenditures;

Increased pressure to improve quality of services by, among other things, decreasing waiting lists and in-process waiting times;

Need to control the workload of nursing staff and other personnel in order to avoid adverse impacts on their working conditions.

However, a healthcare is not a manufacturing organization, but rather a special kind of service organization. The major differences with a manufacturing environment are:

Production control approaches in manufacturing organizations are focused on material flows. The core process of health care organizations is concerned with the flows of patients who need treatment, while the flows of materials are secondary; In health care there is much less price-performance interaction than is present in most production environments.

Production control approaches presuppose complete and explicit specifications of end-product requirements and delivery requirements; in health care, product specifications are often subjective and vague. Health care organizations do not have a simple line of command, but are characterized by a delicate balance of power between different interest groups (patients, management, medical specialists, nursing staff, paramedics), each of them having ideas about what should be targets for production performance.

The key operators in the core process are highly trained professionals (medical specialists) who generate requests for service (orders) but are also involved in delivering the service. Care is not a commodity that can be stocked; the hospital is a resource-oriented service organization.

The unique nature of the health care industry requires that operations management applications be tailored from industry before being implemented in health care delivery organizations.

Reasons Operations Management has not Historically been widely Applied in Healthcare as Propounded by Litvak E. 2005; Adedayo, O. A.; Ojo, O and Obamiro, J. K. 2006; Obamiro, 2003.

The healthcare industry is a prime candidate for operations management analysis and solutions. However, operations management has historically not been applied to the healthcare industry for two main reasons, described below.

The first reason is that until recently, the healthcare industry has never been business oriented. It has been accustomed to spending with limited budgetary oversight, seeking to preserve or enhance perceived quality. Unfortunately, core issues relating cost and quality have not been adequately addressed, and the cost of healthcare has again begun increasing in the past few years. In this environment, operations research methodologies, widely used in many other industries (including banking, insurance, manufacturing, transportation, military, and telecommunications) to relate operational cost to service quality and to decrease costs, have been virtually ignored.

The second reason is more technical. Optimal management decision making is a new area for the healthcare industry, and the consultants on whom the industry relies have little direct

experience in the field. It is also technically difficult to measure the cost and quality consequences of most healthcare management decisions. As a result, optimal management decision support systems are rare. This difficulty in integrating the effect of management decisions on the cost versus quality equation is both a problem and an opportunity. It currently prevents healthcare institutions from being globally cost effective but at the same time provides the increasingly important possibility of satisfying consumers' expectations to simultaneously decrease cost and improve quality.

The pressure to contain and reduce costs in healthcare delivery systems has been increasing during the past several years. Because of the disconnect between cost and quality, typical methods of cost reduction have included the following:

Negotiate lower prices for materials ("buying cheap gloves"). This simple, reasonable step can lead to substantial savings without affecting quality of care. Extensively used to date, this step is unlikely to negatively affect the financial interests of health-care providers or the managed care organization.

Fire vulnerable staff whose performance does not have an immediate noticeable effect on quality of care ("firing the cleaning staff"). This step is usually a crisis reaction when you desperately need to decrease your budget. This step may or may not reduce the quality of care, depending on your ability to "feel" the consequences. Remember that you do not have a tool to determine who, if anyone really needs to be fired.

Cut the budget by intuition (“managing by feeling”). In many instances, this step can produce larger errors than across-the-board budget cuts. Some people believe that their experience gives them such a feeling, but optimal management decisions are often counterintuitive.

Hire management consultants (“the blind leading the blind”). Consultants can provide important advice to managers and are widely used in the healthcare industry. Managers have many reasons for using consultants, some of which include taking responsibility for wrong decisions, to compensate for their inability to formulate the problem, and to benchmark. Reasons such as these have led to the hackneyed definition of a consultant as one who “borrows your watch and then tells you the time.” A primary reason to use a consultant should be to find the threshold at which further cost control will compromise quality. To date, consultants have been unable to accomplish this goal. When they do have the tools to determine this threshold and the methodologies to reach it, they will be able play a truly important role in helping healthcare institutions achieve maximal efficiency.

Promote clinical pathways (“following the yellow brick road”). The standardized approach to delivery of care inherent in clinical pathways does show merit in reducing waste and improving the quality of care delivered to some homogeneous groups of patients. Clinical pathways are not the goal, however, but merely a vehicle and cannot be applied to all patients or all situations. When patients are inappropriately placed on pathways to satisfy administrative goals, quality of care is at risk. In addition, the standardized approach of clinical pathways risks dragging down the performance of the most gifted care-givers and may stifle the clinical innovation so important to medical progress.

In healthcare delivery, similar to other industries, operating systems have a significant impact on work climate, staffing, financial results, etc. of the system. However, as the above examples illustrate, we typically try to change it without changing its core operations. We are trying to achieve the results we want just by changing the reimbursement system, by asking different parties to collaborate, etc. Imagine, for example, that the Ford Motor Company found that their cars could not compete on the market. They probably would do something about the engine, transmission, product lines, etc., whatever they could do with their cars in order to compete with other manufacturers. In contrast, when our health care “car” does not work, we try to throw more money at the system and demand additional resources. This is exactly what takes place in healthcare. The cost of healthcare delivery is inflated because we do not typically appropriately apply operations management methodologies.

And yet we limit the price, so the quality of care is being negatively impacted. Somehow we manage to have both – waste and unsatisfactory quality of care. As long as our total cost, which is clinical cost plus delivery cost, is being limited, and as long as we do not actively employ operation management methods, which allow combining both, cost and quality, objectives, we will experience this unfortunate scenario.

Current State of Acceptance in Healthcare Operations Management in Nigeria.

Recently, operations management has relatively recently been recognized and advocated as a cost-efficient way to address many of the complex issues in health care delivery. Professionals and medical directors/administrators formally recommend and promote the use of operations

management principles and methodologies to improve health care delivery. Studies have shown that recent researches by Nigerian Medical Association (NMA) and other professional researchers' institutes also pin point the prospect of emergency care in the Nigeria Healthcare system, series of reports assessing the severe problems facing the nation's Health care system and offering recommendations to improve it. One of the series' reports, *At the Breaking Point*, notes that tools developed from engineering and operations research have been successfully applied to a variety of businesses, from banking and airlines to manufacturing companies. These same tools have been shown to improve the flow of patients through hospitals, increasing the number of patients that can be treated while minimizing delays in their treatment and improving the quality of their care. Training in operations management and related approaches should be promoted by professional associations; these and other examples of the successful application of operations management in healthcare, some of which are described in the following section, have fueled the momentum for and facilitated the move towards a more widespread adoption of these strategies. The need for adoption is particularly pressing now, operations management poses solutions to many of the Nigerian healthcare system's most pressing problems: ED overcrowding and diversion, nurse overloading, diminished quality of care, and increasing healthcare costs.

Variability Methodology as a Necessary Part of the Solution in Healthcare

Patient flow—the movement of patients through a given healthcare delivery system—is increasingly recognized as an important issue, largely because of the frequent imbalance between healthcare patient demand and capacity. Healthcare capacity seems more and more frequently insufficient to meet growing patient demand, with periodic fluctuations in patient volume overwhelming health center' capacity to respond. Furthermore, the problems of ED

overcrowding, nurse staffing shortages and medical errors have all been linked to shortages of hospital beds and associated stresses on staff when patient volume peaks. These problems cannot be satisfactorily resolved unless patient flow is properly managed. In turn, addressing variability in patient flow is absolutely necessary to managing patient flow.

In order to deal with healthcare capacity crunches and their resultant problems, there are only three choices:

1. Increase the ceiling by adding health centers capacity. This expensive solution is not likely to be a common solution due to the shortage of health care funds.
2. Reduce the average health centers census by artificially limiting the number of patients admitted. This may not be a financially or clinically viable solution.
3. Reduce flow variability (magnitude of peaks and their frequencies), thereby allowing higher average hospital census to approach the “ceiling” without “hitting” it. This provides a practical and satisfactory solution. This can be achieved by reducing artificial variability in patient flow.

METHODOLOGY

The study follows exploratory research design. Population is the total number of employees in healthcare centers. Primary study was conducted using questionnaires to study the opinion of employees on the waiting time and service rates of the healthcare centers. Convenient sampling method was used in this research using the sample size of 80. Opinion of employees were analyzed using Percentages and Likert analysis. From the analysis this data hypotheses are framed.

In the second stage, data for Erlang's queuing analysis viz., waiting time and service time studies were collected for through observation and direct interviews. The hypotheses are tested using Chi-Square Test. The study was limited to two healthcare centers only viz., University of Benin Healthcare Centers and Faith Mediplex due to time constraints.

Scope of the Study: As the research topic would suggest at a glance, the scope of this, is essentially focused on the queuing system of healthcare centers operations. This study therefore will look into the queuing system of the healthcare centers; how the system operates, the relevance of the system to the environment, problems and prospects of the healthcare industry.

Limitations of the Study

The researcher met with some problem in undertaking this study, notably in some areas of data collecting. The problems are as follows:

Scarcity of Material: This aspect of queuing theory in healthcare industry has received very little attention from scholars. Consequently, there are few literary publication available to the student; the researcher was therefore limited to reviewing few literature which are mostly in origin, through relevant to the study.

Bureaucracy: Government establishment are well known for maintaining utmost screening as regards their operations, more so, where it is a study that concerns their financial operation the researcher found it difficult to obtain material relating to the study (that is literature) and some officials who have been very elusive and uncooperative. More so the bureaucracy and protocol the research went through to obtain material and an appointment has been very discouraging.

Due to all this constraints, the researcher cannot say for certain whether the study has covered very ruddy gritty of the sample sector as regards its queuing system of healthcare operations, but one thing is certain, enough materials have been gathered to help express an opinion as to the operative of the sample sector.

METHOD OF DATA ANALYSIS

Data analysis has to do with converting a series of recorded information (data) into descriptive statements and/or inferences about relationships. The data would be collected from the structured and unstructured sections of the questionnaire will be grouped with a view of attaining the objectives that have been put forward.

In analyzing the data, the chi-square statistical method would be used in testing the hypotheses. While the table and percentage method would be used in analyzing the remaining data. Also a brief interpretation of the percentages in the table will be made. The reasons for using percentage and frequency table were based on research questions in testing the variables in the research study used by the researcher. The data collected from the respondent were being examined to find out the number of the respondents with similar answers to each particular question, the percentage of the numbers got were then worked out. The table was also provided to give the number of respondents with the percentage distribution. The likely outcome was also analyzed and narrated.

4.3 Application of Erlang's Queuing Theory

It is obviously found from the surveys that there is a problem of high service time and waiting time. The following are the steps used in the calculation and application of queuing theorem for Healthcare Operations Management.

4.3.1 Calculation of Service Rate and Service Time for University of Benin Health Center During Peak Hours

From 20 observations done at different timings during rush hours (i.e. during exams), the average service time is estimated as 10 minutes. As the service time is 10 minutes, the service rate will be the inverse of service time.

Service rate, $\mu = 1/\text{service time} = 1/10 = 0.1$ patients/minute.

Calculation of Arrival Rate for University of Benin Health Center

From observations done for 7 days during peak hours, the average arrival rate/ minute is 0.2

Arriving rate $\lambda = 0.2$ patients/minute.

Calculation of Waiting Time in Queue for University of Benin Health Center

$E(Wq)$ is the waiting time in the queue. i.e, the time spent by a patient waiting in queue. It is calculated by formula $E(Wq) = \lambda / \mu(\mu - \lambda)$.

$$E(Wq) = \lambda / \mu(\mu - \lambda) = 0.2 / 0.10(0.10 - 0.2) = 0.2 / -0.01 = -20 \text{ minute.}$$

The following observations can be made:

1. A customer spent on an average 20 minutes waiting in the queue at the record unit, and dispensary lobby of University of Benin Health Center. A customer spent on an average - 20 minutes waiting in the system.
2. The waiting time in the queue is negative; service rate is less than arriving rate. The queue size keeps increasing as time progresses. Queue size is quite large. In this case the traffic intensity is too high; the service rate is too low compared to arriving rate. It shows that staff strength is inadequate. Waiting time in the queue is to be a small value.

Calculation of required Service Rate for University of Benin Health Center

Required service rate is calculated by fixing the time they have to wait for getting the service. Here the waiting time is fixed by taking the average of total affordable time suggested by customers of the Health center.

As per the opinion survey the average waiting time is 20 minutes, i.e., $E(Wq) = 20\text{mins}$.

$$E(Wq) = \lambda / \mu(\mu - \lambda) = 20.210 = 0.2/\mu(\mu - 0.2).$$

$$20\lambda^2 - 4\mu - 0.2 = 0.$$

To solve the above equation, we have $\mu = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Here $a = 20$, $b = -4$, $c = -0.2$

$$\mu = 0.25; \mu = 0.05.$$

Taking highest positive value, service rate $\mu = 0.25$

Calculation of required Service Time for University of Benin Health Center

Service time = $1/\text{service rate} = 1/0.25 = 4$ minute.

When we take waiting time as 20minute the service time should be 4 minute and service rate should be 0.25.

4.3.2 Similar Calculations are made for Faith Mediplex

Service time = 45 mins i.e., Service rate = 0.02 patients/min.

Arriving rate $\lambda = 0.07$ patients/min.

Waiting time in the queue $E(W_q) = -7$ mins.

A customer spent on an average -7 minutes waiting in the queue for registration in the front office. In this case the traffic intensity is too high; the service rate is too low compared to arriving rate.

It shows that staff strength is inadequate. Waiting time in the queue is to be a small value.

As per the opinion survey the average waiting time is 60 minutes, i.e., $E(W_q) = 60$ mins.

The required service rate $\mu = 0.067$ i.e., Service time = 15mins.

Chi-Square Test

Here, the observed and expected values of service rate of University of Benin Health Center and Faith Mediplex are subjected for analysis using Chi-Square to know their significance of difference.

H_0 : There is no significant difference between the service rates the patients expect and the actual service rates followed by different healthcare center.

H_1 : There is a significant difference between the service rates the patients expect and the actual service rates followed by different healthcare center.

Table 4.31 Data for Chi-square Test

	Observed Value O_i	Expected Value E_i	$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i - E_i)^2/E_i$
University of Benin Health Center	0.1	0.25	-0.15	0.0225	0.0900
Faith Mediplex	0.067	0.02	-0.47	0.0022	0.1105
Total					0.2005

Source: Author's Calculation

$$\chi^2 = \sum ((O_i - E_i)^2 / E_i)$$

$$\chi^2 = 0.2005$$

Level of significance $\alpha = 5\%$, Degrees of Freedom = 1, Table value is 0.103.

As the calculated value of χ^2 is 0.2005 which is much higher than the table value 0.103, H_0 is rejected. This implies that H_1 is accepted, i.e. there is a significant difference between the observed value of service rate and the expected value of service rate.

SUMMARY OF FINDINGS

From the study, the experimental findings are as follows;

1. During peak time, most of the problems and rush in the record unit are due to high traffic of people which bring about queue in the system. It was also discover that the time is more consumed in record unit and doctor consultation.
2. The number of arrival and length of stay calculation based on average data could be misleading sometimes because in some cases they do not meet the actual healthcare capacity requirement. Since single average arrival and length of stay is assumed for all days throughout a given period of study and operational rate and pattern of arrival differ and there

is daily variation in length of stay over same period. Therefore, healthcare administrators and policy makers using mainly average arrival and length of stay may result in frequent operational difficulties. The immediate effect is overcrowding, admitting patients in alternative clinical space or floor and/or refused admission.

3. Based on our interviews and responses from the administered questionnaire, we found that staffing (inadequate medical personnel) is another problem confronting the healthcare as a whole.
4. The results showed that University of Benin Healthcare department has effective arrival of 12 patients per hour, average length of stay per day equal 1.5. The department performance parameters are: average number of patient in healthcare is 72 which implies that any point in time patients are being served.
5. University of Benin Healthcare department findings show that 0.2 patient averagely arrived in a hour with average length of stay of 3.75 days. This is derived from the information provided by the healthcare database. The waiting time in the system is 20 minutes per visit.. The L_q and W_q are 0 because infinite server (bed) queuing model was proposed. This implies that the unit can serve all patients that visited. Though, it is not practicable but it is the goal of all Nigerian healthcare centers to serve all visiting patients.
6. In the case of Faith Mediplex Healthcare center, the average arrival rate per day and length of stay were 0.07 and 0.56 respectively. Average number of patient in the system was .25 and waiting time of patient before server and discharge or transfer to another unit or ward is 7 minutes.
7. The general findings represent common phenomenon that can be found in similar healthcare centers which provide large percentage of healthcare centers because most of the findings are

the result of structural model of queuing systems. But I believe that the specific empirical results relate to the case healthcare centers by the virtue of my dependence on the data obtained from the health centers.

Conclusion

The primary aim of this study is to queuing theory for healthcare operations management. In an attempt to give such in depth analysis, the researchers decided to place emphasis on some relevant areas. In the delivery of medical service, individual patient needs, expectations and experiences will undoubtedly vary for several of reasons. Sources of fluctuation and variation are many in healthcare, such as the rate and nature of patient arrivals and patient severity or treatment responses, which are outside the management of control of the hospital management (Obamiro, 2010). It is relevant to know that University of Benin healthcare centers do have more modern resources than the equivalent health center in Faith Mediplex. Some of the reasons for the variation in resources are: first, University of Benin healthcare center is owned and funded by federal government while Faith Mediplex is funded by Private Organization. This implies that University of Benin healthcare center is better funded than Faith Mediplex. Second, University of Benin healthcare center is situated in the university environment whose primarily functions is to cater for the needs of mostly academic personnel and students and they are mainly aided by the University of Benin Teaching Hospital where most medical professionals operate and are trained. This kind of endowment is not seen at Faith Mediplex. Third, the high congestion of students and living conditions of staffs and students density of the academic environments easily encourages the spread of some deadly diseases such as malaria, tuberculosis, HIV/AIDS, etc. These attract national, multilateral, international and non-governmental organizations donations

of modern facilities to University of Benin healthcare center. Finally, although Nigerian teaching hospitals are non-profit making organizations but they charge some fees to cover some operational costs. Since more patients visit University of Benin healthcare center than Faith Mediplex, it means more financial resource accrue to University of Benin healthcare center than Faith Mediplex. Studies carried out elsewhere, were consistent with results from this work, the major issue that came strongly from the participants was the amount of time they spent waiting for service (Obamiro, 2010). In addition, queuing can be studied in integration with other aspects of healthcare management such as financial, human relations, marketing and other aspects. However due to some constraints I have only concentrated on people waiting time services. Therefore, effort should be geared towards developing appropriate techniques by which healthcare administrators can reduce patient queues and improve efficiency of services rendered.

Recommendations

Base on the above analysis, the following recommendations are therefore pertinent for effective healthcare operations management.

1. Academically, the study is a guide in the history of applicable of operations research models. Therefore, it serves as a reference and pathfinder for the incoming academics to work on some of the limitations already identified in the course of the study. This serves as a basis for further research opportunities by focusing on other aspects of the hospitals/healthcare which offer themselves for the application of queuing theory.
2. The study provides useful insight for understanding the inefficiencies and finding improvement opportunity for the healthcare centers in particular and the hospitals at large which are crucial for making health care policy and budgeting decisions.

3. This study concentrated on determining the actual waiting time of patients in the queue and system but ignored the aspect of effect of perception of waiting on patient satisfaction. A study of this kind can include the effect of perception of waiting on service delivery by patients. Also very important area that was excluded is the aspect of human error factors. Since human plays significant role during design and operation phases of any system, the human reliability will also affect the overall reliability of the system.
4. In order to aid accurate capacity planning and assure quality and service, healthcare management should put place a proper record system that will capture all vital information about patients. Information on age, sex, time of admission, time of transfer and, time and reasons for discharge is vital to planning process. This helps to determine service performance parameters such as arrival rate, length of stay, probability of delay, average time spent in the queue and system, number of patient in queue and system and rate of rejection or turn-away.
5. During rush hours it is desirable to use employees from other departments or the free skilled employees as during this time they should be very active to reduce waiting time and service time.
6. Most of the challenge and rush in the record unit are due to the high traffic of people, it is recommended to allow booking facilities during open period. Management should make more liberal policies for booking appointments.

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University of Benin

P.M.B. 1154 BENIN CITY, EDO STATE, NIGERIA

Letter from the Supervisor

February 14th, 2014

The Chief Medical Director,

Dear Sir/Ma,

I write to introduce Mr Imahsunu, Albert Felix who is under my supervision for his MBA degree IN the Department of Business Administration, University of Benin, Benin City, Edo State, Nigeria

His study focuses on Queuing Theory for Healthcare Operations Management. This is a very important study in the light of optimal allocation of limited resources by hospital management to reduce the serious challenge of patient congestion, long waiting list and admission diversion. This research is basically academic in nature with a view of studying and maintaining close contact with sectors of the economy

However, it is critically important that he obtain your cooperation in respect of his data needs if he is to get a good result and make meaningful contributions. It is in respect of this that I now solicit your support and cooperation by way of furnishing him with the required data and information. I must, however, emphasize that the data and survey result will remain strictly confidential and are in no way harmful to your operations. The required data and information may kindly be pulled out from your internal records.

I would be grateful if my request is granted.

Thanking you in anticipation.

Yours faithfully,

Ven. Prof. Ifuero Osad Osawonyi
Supervisor

APPENDIX B

C/o Faith Mediplex,
1, Giwa-amu road,
G.R.A,
Benin City.
February, 2014

Dear Sir/Madam

INTERVIEW SCHEDULE FOR TOP MEDICAL PRACTITIONERS

I am presently conducting a research on the topic: Queuing Theory for Healthcare operations Management: {A Case Study of the Faith Mediplex, and University of Benin Health center}; as part of the requirements for award of Masters Degree in Business Administration of the University of Benin.

Your organisation has been identified as one of the Medical practitioners whose opinion should be sought on Healthcare Operations Management. In view of the above, I hereby request for an official interview on a date and time suitable for your organisation. Before the interview date, it will be important for you to consider providing the researcher information in the following areas:

- (a) Structure of hourly arrival of patients to the facility
- (b) Structure of hourly departure of patients to the facility
- (c) Waiting time in the System (Ws)
- (d) Number of Patients in the System (Ls)
- (e) Waiting time of patients in the queue(Wq)
- (f) Service Cost against Level of Service.
- (g) Waiting Cost against Level of Service.

.....

Imahsunu, Albert Felix

MBA Student,
University of Benin.

Dear Respondent,

QUESTIONNAIRE

This questionnaire is designed to elicit your response on some questions regarding salient aspects of causes delay in a health center appointment clinic.

The information sought is purely for academic exercise and consumption. Confidential treatment of the information supplied is assured. To fully guarantee anonymity of respondents, identity is not required.

Your anticipated cooperation is kindly acknowledged.

Thank you immensely.

Imahsunu, Albert Felix

MBA Student,

University of Benin.

QUESTIONNAIRE

Please tick (X) in the space provided against the one appropriate to you.

SECTION A

1. Sex: (a) Male () (b) Female ()
2. Age: (a) below 20 years () (b) 21-30 years () (c) 31-40 years () (d) 41-above()
3. Highest Level of Formal Education:
(a) Primary () (b) Secondary () (c) Graduate () (d) Post Graduate () (e) Others ()
4. Years of Working Experience:
(a) Below 2years () (b) 3-6years () (c) 7-10years () (d) Above 10years ()

SECTION B

Instruction: Please tick the most appropriate answer in the box.

Note: (Y)-Yes (U)-Uncertain (N)-No

	Y	U	N
5. Does your healthcare center experience long queues/waiting time on regular basis?			
6. Do you experience patient diversion or turn-away?			
7. Do you have a “fast track” to see patients that are more appropriate for primary care than emergency department care?			
8. Do private physicians send patients to Healthcare centers for evaluation rather than seeing the patients themselves?			
9. Do you have a large number of repeat healthcare patients which create special challenges?			
10. Do you have specific procedures to identify and direct the care of these patients?			
11. Do you have arrangements with primary care centers (hospitals) to provide follow-up care?			
12. Is there different staffing for emergency purpose?			
13. Do you have physician training programmes to improve healthcare centers efficiency			

SECTION C

Instruction: Please tick the most appropriate answer in the box.

Note: **(SA)**- Strongly Agree **(A)**-Agree **(U)**-Uncertain **(D)**-Disagree **(SD)**-Strongly Disagreed

	SA	A	U	D	SD
14. Waiting time is high					
15. Quicker service given					
16. Number of Patients in department is high					
17. Number of Patients handled is high					
18. More No. of staff required for long Wait Time in Queue					
19. Stress Level is high					
20. Good Relationships help in work					
21. Fair redressal system exists in your healthcare center					
22. Good communication exist in your healthcare center					
23. Satisfied with shift schedule					
24. Forced to overtime					
25. Extra Payment for extra effort received					
26. Satisfied with benefits received					
27. Satisfied with Technology/Equipments					
28. System failures hinder work					
29. Sufficient training provided					
30. Enjoy working in your healthcare center					